

Software Design Description 020

Keyframe Storage and Transmission Protocol

Related Documents

- SDD001 – Image Processing Algorithm: reference frame library concept
 - SDD100 – ESP32-CAM: SD card storage
 - SDD200 – ESP32-DevKitV1: main controller
 - SDD300 – ESP32-S3: receiver and reconstruction
 - SDD400 – FPGA: WHT compression pipeline
 - SDD010 – LoRa link budget and spreading factor
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Overview

Two independent transmission event types exist on the LoRa link. They share the same physical channel but are logically separate and never interleaved.

Type 1 – Scheduled keyframe transmission: Triggered by RTC alarm. Wakes the DevKit from deep sleep. Sends pre-computed WHT coefficient files for all 3 stepper positions over LoRa. S3 stores them in flash. No motion event is associated with this transmission.

Type 2 – Live motion event: Triggered by PIR. DevKit wakes, camera captures, FPGA compresses. Event consists of a preamble packet, a stream of diff coefficient packets, and a post-motion packet. The S3 uses the stored keyframe (selected by stepper position from the preamble) as the reconstruction base.

Keyframe Coefficient Files

The keyframe is stored as WHT coefficients, not raw pixels. A raw 320x240 frame is 76,800 bytes. The coefficient representation is compact enough to transmit over LoRa and store in flash on both ends.

File Naming

Location	Path	Count
CAM SD card	/sdcard/kf_N.coeff	3 (N = 0, 1, 2)
S3 SPIFFS flash	/spiffs/kf_N.coeff	3 (N = 0, 1, 2)

N corresponds to stepper position: 0 = CW (PIR1), 1 = CCW (PIR2), 2 = center (PIR3).

File Format

Identical to the FPGA coefficient output format (blk_schd.v EMIT state):

For each block with at least one nonzero coefficient:

Byte 0: block_row[4:0] | block_col[5:3]

Byte 1: block_col[2:0] | n_coeffs[4:0]

For each coefficient (repeated n_coeffs times):

Byte 0: r[2:0] | c[2:0] | sign | 0

Byte 1: abs(q)[7:0]

Byte 2: (reserved / padding)

End marker: 0x00 0x00

The file is produced by running a static scene frame through the FPGA pipeline once (diff against zeros = raw WHT of the keyframe). The FPGA output is read back by the DevKit via SPI MISO and written to the CAM SD card.

Type 1: Scheduled Keyframe Transmission

Trigger

An RTC alarm wakes the DevKit from deep sleep at a configurable interval (default 4 hours, adjustable via the settings display – see SDD205). The wake mechanism mirrors the PIR interrupt path: RTC alarm asserts a GPIO that the DevKit uses as a wakeup source.

Transmission Flow

```
DevKit wakes from deep sleep (RTC alarm)
|
v
Send START_CAPTURE command to CAM via SPI MISO
CAM rotates to position 0, captures frame, runs FPGA pipeline
FPGA produces kf_0.coeff -> DevKit reads via SPI MISO
CAM writes kf_0.coeff to SD (/sdcard/kf_0.coeff)
DevKit sends kf_0.coeff over LoRa (0xAA 0x03 packets, position=0)
|
Repeat for positions 1 and 2
|
DevKit returns to deep sleep
```

LoRa Packet Format (Type 1)

Keyframe packets use magic 0xAA 0x03. Each LoRa packet carries a chunk of the coefficient file. Multiple packets may be needed per position (RYLR998 max payload = 240 bytes).

```
Byte 0: 0xAA
Byte 1: 0x03 (keyframe packet type)
Byte 2: position index (0, 1, or 2)
Byte 3: chunk sequence number (0-based)
Byte 4: total chunks for this position
Byte 5+: coefficient data (up to 235 bytes)
```

S3 reassembles chunks per position, writes complete file to SPIFFS on receipt of the final chunk.

Type 2: Live Motion Event

Preamble Packet (0xAA 0x01)

Sent before any diff data. Tells S3 which keyframe to load and provides environmental context.

```
typedef struct {
    uint8_t magic[2];           /* 0xAA 0x01 */
    uint8_t kf_idx;             /* stepper position 0-2 -- selects keyframe file */
    uint8_t stepper_phase;      /* raw motor phase */
    uint32_t timestamp_s;       /* seconds since boot */
    int16_t temp_x10;           /* tenths of Celsius */
    uint16_t humidity_x10;      /* tenths of %RH */
} __attribute__((packed)) event_start_t;
```

On receipt, S3 loads /spiffs/kf_N.coeff (where N = kf_idx) into PSRAM as the reconstruction base for this event.

Diff Packets (0xAA 0x04)

Each packet carries one or more compressed block payloads from the FPGA output stream. Format TBD – see SDD401 for the EMIT byte layout.

Post-Motion Packet (0xAA 0x02)

```
typedef struct {  
    uint8_t  magic[2];          /* 0xAA 0x02 */  
    uint32_t timestamp_s;       /* event end time */  
} __attribute__((packed)) event_end_t;
```

S3 Keyframe Storage

The S3 stores 3 keyframe coefficient files in SPIFFS. At startup, all 3 are loaded into PSRAM (one per position). On motion event preamble receipt, the matching position's coefficients are used as the inverse-WHT reconstruction base.

PSRAM Layout (S3)

Region	Content	Size
psram[0]	Reconstructed frame buffer, position 0	76,800 B
psram[1]	Reconstructed frame buffer, position 1	76,800 B
psram[2]	Reconstructed frame buffer, position 2	76,800 B
psram[3]	Current event working buffer	76,800 B

Keyframe coefficients themselves stay in SPIFFS (flash). They are applied at reconstruction time, not pre-expanded to pixel buffers.

Recent Event Storage (S3)

The most recent complete motion event is stored in SPIFFS as a binary log: `/spiffs/last_event.bin`.

Format: raw concatenation of all received LoRa packets for the event (preamble + diff packets + post-motion), in receive order. The Flutter app can request this file via WebSocket and replay it through the same reconstruction path used for live streaming. User is presented with delete/keep options; files older than a configurable TTL are purged on boot.

Packet Type Summary

Magic	Type	Direction	Purpose
0xAA 0x01	event_start_t	DevKit -> S3	Motion event preamble
0xAA 0x02	event_end_t	DevKit -> S3	Motion event end
0xAA 0x03	keyframe chunk	DevKit -> S3	Scheduled keyframe data
0xAA 0x04	diff data	DevKit -> S3	Motion diff coefficients

Implementation Status

PLANNED – not yet implemented.

Component	Status
kf_N.coeff write on CAM SD	Pending
kf_N.coeff write on S3 SPIFFS	Pending
RTC wakeup for keyframe transmission	Pending

Component	Status
LoRa keyframe chunk protocol (0xAA 0x03)	Pending
S3 keyframe chunk reassembly	Pending
S3 last_event.bin storage	Pending
Flutter app event replay / download	Pending
FPGA pipeline verified (Bug 1/2/3 fixes)	In progress